

Managing Technology-Based Projects in Multinational Environments

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Abstract—This paper examines the managerial issues involved in executing technology-based projects that span international borders. The results of a bilateral comparative field study, conducted in Brazil and the United States, show that in spite of considerable social and cultural differences between two international operating environments, successful integration of multinational projects may not require fundamentally different management approaches. This, however, does require strong senior management support and efforts in developing effective organizational linkages and alliances. Managers must need to have focus on cross-boundary relationships and fine-tune their people skills to deal effectively with complex forms of negotiations, delegations, and commitments in systems which are often weak on formal command and control.

Index Terms—Cultural diversity, management style, multinational, project integration, project management, project success, technology.

I. INTRODUCTION

ORGANIZATIONS that manage projects across international borders often find it frustratingly difficult to implement their well-thought-out plans. These multinational projects need to be integrated not only across different geographic regions, but they also need to be unified among different business processes, management styles, operational support systems, and organizational cultures [4], [20], [22]. While the managerial issues and challenges of cross-functional integration and project leadership have been known for a long time and have been the focus of extensive studies, little has been published on the effectiveness of managerial leadership styles across geographic and cultural regions, which is the focus of this study. Such cross-cultural management practices are quite common in today's global business environment with multinational sourcing, joint ventures, and alliances. While unified project direction toward technology transfer, integration, and business objectives usually comes from one central location, they need to be effectively communicated and "managed" across international lines and organizational cultures [42].

A. Multinational Project Environment

Project environments which extend internationally exhibit intricate and enormously diverse characteristics in type, com-

plexity, technology, and customer orientation. Managers have to deal with differences in languages, time zones, organizational, and personal cultures, policies, regulations, business process, and political climate [2], [12], [34], [49]. As a result of these complexities, new alliances are often being formed among companies and new concepts of technology transfer evolve, such as concurrent engineering and process action teams. Further, these more dynamic and intricate conditions also call for more sophisticated management skills and project leadership [65].

B. Project Complexity

Yet another managerial challenge is the complexity of today's projects and their support environments. Technology has become a significant factor for almost every business, and it affects project activities from small to large businesses and from private industry to government and aerospace. When describing their operations, whether product, process, or service oriented, managers point to specific indicators of project complexity. They mention the high degree of technical complexities [11], [21], evolving solutions [1], [5], [20], high levels of innovation and creativity, technology transfers well integrated with the business processes [10], [26], [34], [66], multidisciplinary teamwork and decision making [18], [21], [29], complex support systems such as computer-aided design (CAD), computer-aided manufacturing (CAM), enterprise resource planning (ERP), and design for manufacture and assembly (DFM/A), intricate multicompany alliances [16], [20], [25], [44], and highly complex forms of work integration [49], [58], [59], [61], [67], [72]. In addition, today's projects and their integration rely to a considerable extent on member-generated performance norms and evaluations, rather than hierarchical guidelines, policies, and procedures [4], [28], [31], [33], [41]. As a result, self-directed teams are gradually replacing traditional, more hierarchically structured project teams in effectively orchestrating and controlling complex projects [34], [62], [68], [69]. This paradigm shift is the result of changing organizational complexities, demands, and cultures. While this shift is an ongoing and gradual process, at some point it requires a radical departure from traditional management philosophy based on organizational structure, motivation, and leadership. Effective role performance in such a contemporary work environment requires a more sophisticated management style which relies strongly on group interaction, resource and power sharing, individual accountability, commitment, conflict handling, cross-functional linkages and cooperation, technology transfer models, top management involvement, self-directed teams, and design/build approaches [4], [30],

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[59], [65]. As a result of these shifts, traditional management tools, designed largely for top-down control and centralized command and communications, are no longer effective and stratified models of leadership do not fit all situations [13], [15], [24], [29], [38], [67].

Traditionally, research on the determinants of project management performance has focused either on the administrative process of project planning, tracking and controlling, or on the behavioral aspects of motivation, team building, and leadership [63]. However, in recent years an increasing number of studies have broadened the investigation into other areas. Research published by Asakawa [2], Keller and Kedia [34], and Thamhain [63] shows the enormous intricacy of variables and concepts involved in the successful management of today's multinational projects, especially when they involve complex technology issues. Therefore, it is not surprising that contemporary managers often applaud writers such as Drucker [23], Moss Kanter [48], Peters [51], Quinn [53], and Roberts [54], who emphasize the nonlinear, intricate, and often messy processes of multifunctional teamwork that have become the new paradigm for organizing and executing complex and technology-based projects. All of these challenges are areas of major concern to managers in project-oriented business environments. They also represent the critical functions which determine success or failure in our complex markets.

II. CONDITIONS CRITICAL TO PROJECT PERFORMANCE

Over a period of five years, we have examined over 80 factors which were suggested by managers as critical to high project performance [39], [63], [65]. The factors which correlated strongest were grouped into eight "classes" of conditions which seem to have a strongly favorable influence on project performance. The process of defining, testing, and grouping these factors is discussed in more detail in the Appendix of this paper. Furthermore, these eight classes of factors seem to have a favorable impact on a number of conditions which are commonly recognized as desirable for high project performance, such as effective teamwork, multifunctional support, innovation, quality, personal enthusiasm, and commitment [39]. We have labeled these eight sets of measures, which are critical to project success, managerial conditioning practices (MCP's). They are listed below and briefly described in Table I.

MCP1	Project leadership.
MCP2	Work design and delegation.
MCP3	Management support.
MCP4	Communications.
MCP5	Work challenge.
MCP6	Personal drive and motivation.
MCP7	Minimum conflict, risk, and threats.
MCP8	Personal appraisals and awards.

While we recognize that there possibly are many other measures associated with project performance in addition to the 80 factors examined earlier, the eight sets of measures grouped into MCP's seem to have the greatest influence on project performance. They also have the greatest impact on success in all project situations and in all countries which we studied and were also sustained over multiyear periods and over a large spectrum

of different projects and host companies. As part of this broad and complex situation, we also suspect that the organizational environment might have a considerable influence on the managerial conditioning practices. Specifically, we realized from previous studies that the business environment affects the way management leads, directs, and supports these projects. Thus certain managerial practices may be seen as more important in one company than another. This is especially evident for those projects that are jointly managed across international lines and show considerable differences in infrastructure, organizational, and personal culture. Therefore, managerial practices which are effective in one organization may not be effective in another. The underlying issues and implications will be examined in the current field investigation. The findings could have a significant impact on the management of multinational project management where management directions are often uniformly applied across different countries and organizational cultures.

III. OBJECTIVE, SCOPE, AND SIGNIFICANCE

A bilateral field study was designed and conducted to measure the similarities and differences of operational conditions and managerial practices in different project environments across international borders. The objective was to gain insight into: 1) the effectiveness of management practices involving multinational projects and 2) possible methods for strengthening the managerial process involving multinational project integration. Consistent with prior research work by the authors, this study focuses on complex and often technology-intensive project environments.

The significance of examining these similarities is in several areas. First, the study could help in determining the degree to which managerial styles, working conditions, project tools, techniques, processes, policies, and procedures are transferable from one project to another or from one company to another. Hence, the study could help to determine to what degree the lessons learned in one situation apply to another. Second, we could develop insight into the differences and/or similarities of management practices between such geographically, culturally, and politically different regions as Brazil and United States. This would also provide the potential opportunity for extending the findings into other regions. Third, the study might be important for multinational projects which involve highly diverse cross-functional teams which must be coordinated and integrated across geographic and cultural lines. Finally, the study provides a methodological framework for further research and the opportunity to build a data base for comparative organizational studies.

IV. METHOD

This exploratory field study is the extension of ongoing research into best practices of managing complex, technology-oriented projects, with regularly published results [37]–[39], [69]. Data were collected between 1993 and 1996 from 125 Brazilian and 95 U.S. project leaders, their team members, and superiors who shared their experiences of managing multinational projects. These teams consisted mostly of technical personnel, such as engineers, scientists, and technicians, involved

TABLE I
MANAGERIAL CONDITIONING PRACTICES CRITICAL TO PROJECT PERFORMANCE

CRITICAL FUNCTIONS (Managerial Conditioning Practices) MCP	DESCRIPTION
MCP 1: Leadership	Effective direction and leadership may be required at all levels of the project organization throughout the project life cycle. This includes project definition, organization, task integration, group decision making, problem identification, and search for technical solutions and conflict resolutions.
MCP 2: Work Design and Delegation	The roles, responsibilities, expectations and evaluations must be clearly defined at the beginning. Similar to the indoctrination of a new employee to a company, a new project team member must be "signed-on". This should include a one-on-one discussion with the project leader regarding the specific assignment, its overall objectives, timing, budgets, interfaces, the specific role of the new member within the total project team, and the type of performance evaluation to be expected.
MCP 3: Management Support	Visible upper management involvement and support to the project team, including the commitment to agreed-on resources, is often important to maintain team focus, motivation and commitment to desired project results.
MCP 4: Communications	Successful project teamwork may require effective communications among team members and with project support organizations. Such communications involve many forms of oral, written, and data-based transfer of information, via a great variety of channels, both formal and informal in nature.
MCP 5: Work Challenge	In spite of rigid project plans with predefined objectives, scopes, and resources, project leaders often have the flexibility to accommodate the professional interests and desires of support personnel when negotiating and assigning individual tasks.
MCP 6: Personal Drive and Motivation	Project managers can influence the climate of the work environment by their own actions. Concern for project team members and the ability to integrate personal goals and needs with the project objectives, can stimulate personal enthusiasm and a climate high in motivation, work involvement, open communications, and ultimately high project performance.
MCP 7: Minimum Threats and Conflict	Project leaders must foster a work environment which is low on personal conflict power struggles, surprises, and unrealistic demands. Risk taking should be encouraged to some degree, and failures of well-intentioned efforts require special managerial support. An atmosphere of mutual trust and respect is necessary for project personnel to communicate problems and concerns candidly early in their developments.
MCP 8: Personal Appraisals and Awards	Project team members should be clear on <i>how and by whom</i> their professional performance is being measured, and in what form it affects their organizational rewards.

in technology-oriented project work such as new product developments, R&D, and field service projects. These teams were employed by large technology-based multinational companies of the *Fortune 1000* category. Together, the data covered 220 projects with budgets averaging \$752 000 and durations averaging 12.1 months. However, the current study was not designed to investigate possible differences or similarities among the various projects types. Data were collected by questionnaires, participant observation, and in-depth retrospective interviewing.¹ The purpose of this combined data collection method was to cast

¹These methods have been used by the authors and others in similar research applications. A more detailed description of this data collection method is given by Thamhain [63].

the broadest possible information-gathering net to identify the similarities and differences of managerial practices between the two operating environments of Brazil and the United States. The questionnaire was designed to measure the perceived: 1) importance and 2) effectiveness of each of the eight classes of MCP's. To minimize the potential biases which might have resulted from the use of social science jargon, eight specific statements were developed, similar to those in the description column of Table I, to define each of the eight management practices for the questionnaire. Managers were asked to characterize their management style and work environment by judging each statement (MCP) as to: 1) the importance of the management practice to their performance and 2) the effectiveness of the management

TABLE II
PERCEIVED IMPORTANCE AND EFFECTIVENESS OF MANAGERIAL CONDITIONING PRACTICES

CRITICAL FUNCTIONS (Managerial Conditioning Practices) MCP	IMPORTANCE TO SUCCESSFUL PROJECT MANAGEMENT AS SEEN BY				EFFECTIVENESS OF USE IN PROJECT MANAGEMENT AS SEEN BY			
	Senior Managers		Project Leaders		Senior Managers		Project Leaders	
	Brazil	USA	Brazil	USA	Brazil	USA	Brazil	USA
MCP 1: Leadership	4.72	4.65	4.68	4.41	3.85	3.55	3.75	4.22
MCP 2: Work Design & Delegation	4.63	4.25	4.65	4.21	3.88	3.32	3.78	4.07
MCP 3: Management Support	4.68	3.88	4.54	4.21	3.65	4.15	3.25	3.02
MCP 4: Communications	4.50	4.38	4.53	4.57	3.58	3.41	3.65	4.23
MCP 5: Work Challenge	4.28	3.50	4.28	3.80	3.40	3.33	3.45	3.22
MCP 6: Personal Drive, Motivation	4.35	4.45	4.15	4.40	3.58	4.05	3.56	4.39
MCP 7: Minimum Threats, Conflict	4.21	3.05	4.22	4.03	3.45	2.38	3.25	2.27
MCP 8: Appraisals & Awards	4.18	3.72	4.18	3.32	2.80	3.22	2.80	2.51

practice used. Both, the importance and effectiveness were measured on a five-point Likert-type scale: 1) strong disagreement; 2) disagreement; 3) neutral or some agreement; 4) agreement; and 5) strong agreement.

Standard statistical methods were used to summarize the survey data such as shown later in Table II. The agreement among the various populations was tested using Kruskal–Wallis analysis of variance by ranks. Further, the association among the various sets of variables was measured using nonparametric Kendall’s Tau rank-order correlation.

While the bilateral design of this study is a recognized limitation, it is also an opportunity for future research. Identifying similarities and differences in management practices among project leaders of two countries is one facet of studying the social and cultural distinctions that characterize multinational project organizations.

V. RESULTS

The results of this field investigation are presented in three parts. First, the relative importance and effectiveness of managerial conditioning practices is summarized, comparing Brazilian and U.S. leadership styles. Second, the criteria for effective leadership in multinational project environments are discussed. Finally, the lessons learned from this field study will be discussed regarding effective multinational project management.

A. Importance and Effectiveness of Managerial Practices

Table II summarizes the importance and effectiveness of eight managerial conditioning practices as perceived and ranked by Brazilian and U.S. senior managers. The results show that on average both Brazilian and U.S. managers see all conditioning practices as important and effective.² However,

²The findings are consistent with earlier studies [39], [65] which confirmed the importance of all eight MCP’s for high project performance. Additional insight is now provided via comparison of perceived MCP importance and effectiveness between managers in Brazil and the United States.

it is interesting to note that traditional managerial functions, such as leadership (MCP-1), work design (MCP-2), management support (MCP-3), and communications (MCP-4), were perceived as more important and more effectively practiced than contemporary functions, such as work challenge (MCP-5), personal drive and motivation (MCP-6), minimum threat and conflict (MCP-7), and personal appraisals and awards (MCP-8). This is particularly true in the case of Brazilian senior managers and project leaders. In project-oriented work environments, managers seem to feel that extrinsic motivation via organizational infrastructure must be provided first before more intrinsic motivation and influences can be effective and beneficial toward project performance.

B. The Importance-Effectiveness Gap

The data in Table II also reveal that, in both countries, importance rates higher than effectiveness. Both the project leaders and their superiors perceive the importance of each of the eight MCP’s higher than the effectiveness of using them. On average, the perceived importance ranks one full point higher (out of a five-point measurement) than effectiveness. Managers recognize the high level of importance of these conditioning practices (MCP’s) while acknowledging the difficulties of effective implementation. The gap between “importance” and “effectiveness” exists in both the Brazilian and U.S. environment. Brazilian managers, however, indicate a more consistent differential of about 1.0 point, while U.S. managers seem to be less uniform in their disagreement, which is shown by the great variance of the data, especially among senior managers.

However, in spite of the gap between the two measures, a significant agreement exists between the populations of project leaders and their superiors as to the degree of MCP importance and effectiveness rating, which was confirmed via Kruskal–Wallis analysis of variance by rank at a confidence level of 97% ($p = 0.03$). A similar agreement was measured between the sample population of Brazil and the United States, however, with a somewhat lower confidence level of 94%

($p = 0.06$). The gap between importance and effectiveness represents some form of performance measure. This is expressed in Table III as the ratio of “perceived effectiveness to perceived importance” and is called “performance factor.” It represents a “quick measure” of normalized managerial effectiveness which can be used to compare and benchmark managerial conditioning practices and work environments.

C. Senior Managers Versus Project Leaders

In comparing the two groups of: 1) senior managers and 2) project leaders, we find a very close agreement in the Brazilian environment between importance and effectiveness of virtually all MCP's, while in the United States the data show clear differences between the two measures, as seen in Table II. A more specific analysis is provided below.

1) *Importance of Managerial Conditioning:* For the Brazilian sample, both groups of: 1) senior managers and 2) project leaders agree reasonably closely³ with each other on all MCP's. In the United States, the two groups agree on the level of importance for most MCP's. There are, however, some noticeable exceptions: MCP-3 (management support) and MCP-7 (personal conflict, threat, power struggle, and trust) were perceived at a higher level of importance by project leaders,⁴ while MCP-8 (personal appraisals and awards) was seen as lower in importance by the project leaders than by their senior managers.

2) *Effectiveness of Managerial Conditioning:* Senior managers and project leaders in Brazil agree again relatively closely on all MCP's, with the exception of management support (MCP-3), which was considered lower for project leaders than for their senior managers. In the U.S. data, differences between the two managerial groups are even more pronounced. Functions which are mostly under the control of senior managers, such as management support (MCP-3) and personal appraisals and awards (MCP-8), were perceived as more effectively practiced by senior managers than by project leaders, while functions under the control of project leaders, such as project leadership (MCP-1), work design and delegation (MCP-2), and communications (MCP-4), were seen at a higher level of effectiveness by project leaders. While the differences in U.S. environments could be explained via attribution biases, this does not explain the lack of such differences among Brazilian managers, which could have significant implications for multinational project management. Yet, in spite of the specific differences discussed above, some principle agreement exists between the two groups of senior managers and project leaders, as we can see from examining the data in Table II. Two specific statistical tests were used to quantify this level of agreement for future comparison and research. Kendall's coefficient of concordance yields $W_B = 0.94$ for the agreement among Brazilian managers and $W_{U.S.} = 0.90$ for Americans. The

TABLE III
PERFORMANCE FACTORS IN PROJECT MANAGEMENT

CRITICAL FUNCTIONS (Managerial Conditioning Practices) (MCP)	PERFORMANCE FACTORS	
	Effectiveness Importance	
	Brazil	USA
MCP 1: Leadership	.82	.86
MCP 2: Work Design & Delegation	.83	.87
MCP 3: Managmt Support	.75	.89
MCP 4: Communications	.82	.85
MCP 5: Work Challenge	.79	.90
MCP 6: Personal Drive, Motivation	.83	.95
MCP 7: Minimum Threats, Conflict	.79	.65
MCP 8: Personal Appraisals & Awards	.67	.81

Kruskal-Wallis analysis of variance by ranks shows at a 95% confidence level that both the senior managers and project leaders come from populations with similar distributions.⁵ Therefore, we can conclude that there is no statistically significant difference in the way the two managerial groups see the relative importance and effectiveness of the managerial conditioning practices.

D. Brazil Versus the United States

On average, senior functional managers and project leaders rate MCP's in the range of “important” to “very important” (the average is “very important”) and also indicate by and large an “effective use” of these managerial practices. This indicates that both groups of managers, in both countries, perceive these managerial practices as critical for completing their projects according to established resource and quality plans. At the same time they feel reasonably confident about the managerial process. Yet further analysis will provide additional insight into the dynamics and intricacies of multinational project management.

While the perceptual differences between the two groups of senior resource/functional managers and project leaders were small in each country (as shown in Table II), there are some interesting differences in managerial thought and practice between the two countries with implications for multinational project management. First, as can be seen from the data in Table II, the managerial perception of MCP importance and effectiveness varies over a wider range in the United States than in Brazil, indicating a less uniform agreement among U.S. managers across the spectrum of managerial practices. Second, the gap between perceived importance and actual MCP practice is wider in Brazil than in the United States. Further, the statistical significance of the differences between Brazil and the

³This level of agreement was quantified via Kruskal-Wallis analysis of variance by ranks at a 95% confidence level. That is, both the senior managers' and the project leaders' scores come from populations with similar distributions.

⁴Management support (MCP-3) is 0.33 points more important to project leaders than to senior managers. Work environments with low levels of personal conflict, power struggle, threats, and high trust (MCP-7) are 1.0 point more important to project leaders than to senior managers.

⁵The same confidence level was calculated for both Brazilian and U.S. samples within a 0.6-point range, and is therefore not separately quoted in the text.

United States was tested by using the Kruskal–Wallis analysis of variance by ranks which shows at a 97% confidence level ($p = 0.03$) that the two samples follow indeed a distribution.

The significance of these findings is in the area of multinational project operations. The differences of MCP effectiveness ratings among managers suggests differences in managerial cultures, values, and operational policies. The specific data from this field investigation and its metrics could be used for comparative studies, establishing building blocks for a theoretical framework and more insight into the complex workings of multinational project management processes and their interfaces. Furthermore, it could establish the basis for designing work processes and managerial procedures for multinational project activities, including channels of communications, decision making, and levels of autonomy.

VI. IMPLICATIONS ON MANAGERIAL PERFORMANCE

An integrated statistical measure can be derived from the project environment with implications for managerial performance. For each MCP, we defined a performance factor as the ratio of perceived effectiveness to perceived importance. This ratio indicates how effective each critical function is managed relative to its importance. A ratio below 1.0 indicates a potentially unfavorable situation. The smaller this ratio, the less effective is a particular managerial practice used within the organization, relative to its importance. Table III summarizes these ratios as performance factors across all MCP's for the four groups of senior managers and project leaders in both Brazil and the United States. On average, Brazilian performance factors are lower than U.S. factors. This could indicate that Brazilian managers, on average and relative to their U.S. colleagues, see themselves as somewhat less effective in conditioning their teams and work environments, or see the importance of this conditioning at a higher level, or Brazilian managers are just more critical and conservative in assessing these managerial practices and their effectiveness. In either case, the data indicate a small but statistically significant difference ($p = 0.04$) between the two groups. While the reasons could be differences in managerial style, team characteristics, and work environment, these implications are important for multinational team building. The findings suggest that the differences in cultural impedances between project teams separated by national borders might be significant enough to impact the effectiveness of managerial leadership style from one team to another or from one central control office to a local team. This would suggest that "local" leaders are more effective in managing the local team, rather than in unifying project teams at-large across international boundaries, which is a proposition that should be further researched.

The data further show a significant gap between senior managers and project leaders in U.S. project environments, while Brazilian managers exhibit a relatively strong agreement. While these findings validate the results of the previous section, analyzing Brazil versus the United States, they also provide some new insight into leadership style. As can be seen from a comparison of performance factors of senior managers and project leaders, each group has the most favorable perception of practices they control but a less favorable perception of

practices controlled by the other group. Specifically, the U.S. project leaders show favorable factors on: 1) project leadership; 2) work design and delegation; 3) project communications and motivational practices, while senior managers score lower on these ratios. On the other hand, practices related to: 1) management support; 2) conflict resolutions; and 3) personnel appraisals and rewards appear more favorable for senior managers in comparison to project leaders.

While it might be tempting to use attribution concepts in explaining the polarization of the two U.S. groups, it would not be consistent with the actual observations of relatively unified Brazilian groups, which suggests a more complex and subtle situation. What we can glean from the analysis is some insight into the diverse managerial thoughts and possibly diverse leadership styles of culturally different regions, which explains in part the difficulties project managers experience in trying to establish a common project management process and an unified framework of direction and leadership in a multinational environment. Furthermore, the significance of these findings is in the area of managerial skill development and organizational support, both of which were mentioned by managers during follow-up interviews. They are seen as crucial for effective managerial role performance and for resource decisions on training and organizational development issues.

VII. LESSONS LEARNED FOR MULTINATIONAL PROJECT MANAGEMENT

As a multidisciplinary field, project management has always been concerned with changes in managerial practices and business processes. However, with the globalization of today's industry, the successful implementation of project plans depends not only on the effective use of managerial tools and techniques in one particular organizational environment but, equally important, on the effective use of these techniques across different geographic regions which often incorporate great differences in their organizational cultures. For many companies, defining and implementing an effective process for managing multinational projects amounts to managing organizational change with all of its dynamics, complexities, risks, and challenges. Thus, one of the greatest challenges is to adopt continuously changing management techniques to local cultures and organizational values without losing consistency, purpose, and managerial integrity.

Yet from a different organizational perspective, successful multinational project management has to involve the ability to build a unified managerial process for effective technology transfer and integration. Such a process must have enough flexibility to adapt to local leadership while functioning consistently within established organizational norms and cultures. The real challenge for companies engaged in multinational project management lies not so much in identifying these barriers but in getting the organization to recognize their significant impact on business performance and to invest resources for optimizing the multinational business process.

As modern project management becomes more oriented toward self-directed teams, more flexible, and less hierarchical, with controls based to a large extent on commitment, motivation, and local team leadership, there appears a great opportunity

for designing multinational project management systems which accommodate both the local organizational environment and its multinational superstructure necessary for overall project integration and control. However, building such multitier processes requires more than simply empowering people at the local level. Senior management needs to work with their organizations to build strong linkages between the local project teams, their support systems, and the overall project organization and its leadership. These partnerships are more readily achieved when managers in all organizations sense a strong degree of cross organizational dependency. This requires an understanding of the technical and managerial challenges involved in the project. It also requires personal commitment from these managers to build the multifunctional and cross-cultural people alliances necessary for implementing the multinational project plan.

Finally, the findings have implications on managerial power and influence. Given the political nature of organizations, the diversity of organizational culture, and the possible regional differences of managerial style examined in this study, it appears that power is often shared between managers of the local organization and the project office or sponsor organization. In some areas there may be even a power vacuum, that is, jurisdiction, authority, or responsibility over certain issues and decisions may not be clear. This provides an opportunity for managers to enlarge their sphere of power and influence. While such shifts in organizational power and influence are natural and predictable, they do not necessarily enhance the business process. Usually, such shifts in power and influence are by-products of organizational change and warning signs that the managerial process is changing and requires fine tuning. While such changes could have either beneficial or detrimental impacts on project performance, they should be examined and dealt with to avoid the risk of organizational tension, mistrust, conflict, and power struggle.

VIII. CONCLUSION

Managing projects across international borders is difficult and challenging. This paper has explored the differences, similarities, and challenges of managerial practices encountered within such multinational project operations. As we move toward the next millennium, the opportunities for multinational project ventures are increasing, exerting pressures on businesses to exploit them in spite the challenges. Hence, for companies to compete effectively in our global economy, managers must be able to lead and unify project teams successfully across international lines. As projects become more multinational they also grow more complex in both their organizational and leadership dimensions. Managing these projects requires highly sophisticated business processes and people skills to ensure effective technology transfer and integration. This study illuminates the enormous complexities involved in managing projects across international lines which, in part, seem to result from cultural diversity and differences in organizational process, norms, and procedures. In addition, different types of projects, including size and duration, might influence the situational effectiveness of managerial leadership style in a multicultural environment. Clearly all of these issues suggest important areas for future research. Yet in spite of the broad

scope and challenges of managing multinational projects, the managerial processes of defining, organizing, and integrating the project toward its desired results is very similar to projects limited to domestic boundaries. What is different relates to power, control, and politics at the local level, which ultimately affects the integration of the project and unified decision making across organizational boundaries. Management is often too involved with the challenges of running the business to be concerned with the subtle differences, conflicts, and operational complications which are natural and should be expected when working across culturally different environments. As a result, organizational interfaces are not always developed for the effective cross-functional and multicultural cooperation necessary for achieving the integration of complex projects across multinational boundaries. The management of either partner organization should not expect, by simply issuing a work order or project summary plan, to create a unified project team, with the ability to work seamless across organizational lines and international borders. Management must recognize the greater autonomy of their partner organizations as well as their cultural differences. They must build a true partnership among all the contributing organizations with strong linkages for communication, decision making, and technology transfer. Such a partnership is most likely to evolve if all team members throughout the project organization share the same objectives and commitments to desired results. Emphasis on common values and goals, rather than on differences, will help to focus and unify the team. Moreover, these multinational alliances must not only be built at the beginning of the project venture, but they must be continuously refueled and maintained over the lifetime of the project. Both top management and the project leaders in the various partner organizations can help in developing such effective multinational project organizations.

Information technology provides a powerful set of tools and techniques in support of multinational project operations. Multimedia conference calls, e-mail, fax, groupware, and Internet websites are just a few examples of available information technology that supports today's project managers in communicating, data sharing, and integrated decision making. These tools have the ability to link organizations across geographic regions and time zones. They also help to partially neutralize cultural differences. The growing arsenal of standard hardware and software not only helps to administer the project and manage the technical challenges, but it also helps to reduce cultural barriers and to increase team cohesiveness without compromising the manageability at the project team level.

One of the interesting findings of this study is that managing these multinational projects effectively may not require a fundamentally different approach from that needed to succeed in domestic environments. What is needed is a fine tuning of the business process to strengthen organizational linkages and a managerial awareness of the cultural differences among their linking organizations by all key players. Sophisticated people skills are crucial to effective role performance. Project managers must cross organizational, national, and cultural boundaries and work with people over whom they have little or no formal control. To function effectively, these managers need to build alliances and develop collaborative ventures. All of this

requires intense focus on cross-boundary relationships, negotiations, delegation, and commitment, in addition to establishing formal command and control systems at the local level. To be effective in such a multinational team environment, the manager must be a social architect who understands the interaction of organizational and behavioral variables. This understanding will facilitate a climate of active participation, minimum dysfunctional conflict, and effective communication. It will also foster an ambiance conducive to change, commitment, and self-direction. This awareness has fostered a growing emphasis among business leaders to find better ways of integrating managerial practices with business strategy. Management development efforts often play an important role in fine tuning management practices, leadership style, and business process. When properly orchestrated among global partner organizations, such management development can help to unify the project organization. While no single set of tools or broad guidelines exists that guarantees instant managerial success, the understanding of organizational variables and their interrelationships to the behavior of people helps managers to examine and fine tune their leadership styles, actions, and resource allocations toward continuing organizational improvement. We hope that the findings will stimulate further managerial thought and insight for both management scholars and practitioners toward more comprehensive models and management tools in order to achieve the high performance needed in today's technology-oriented multinational project environments.

APPENDIX

MANAGERIAL CONDITIONING PRACTICES—HOW THE MEASURES WERE DEVELOPED

Several exploratory field studies were conducted by the authors in technology-based work environments to determine the factors which significantly influence project performance. These studies were conducted in various countries, including Brazil and the United States, covering a five-year period from 1991 to 1996. Using initially a questionnaire and interview-based delphi approach, subsets of leadership style, work environment, project management support system (including tools and techniques), team and team member characteristics, and project metrics were identified. These investigations resulted in the identification of 83 variables with potential influence on project performance. These variables were then used to characterize or "measure" project environments as a perception of their team members. Also, project performance was measured as a perception of senior management. As the next step, a correlation analysis was performed on each of the 83 variables to measure their association with project performance. Nonparametric statistics, such as Kendall-Tau and Komogorov-Smirnov, were used to calculate the associations between each project variable and overall project performance and their significance. These studies led to 25 broad conclusions, such as:

The more interesting and professionally stimulating the work is perceived by team members, the higher is the project performance and overall degree of success being perceived by senior management.

1) *Short-Form Conclusion:* Interesting work has a favorable influence on project performance.

The more cross-functional involvement was generated during the project definition phase, the higher is the project performance and overall degree of success being perceived by senior management.

2) *Short-Form Conclusion:* Cross-functional participation during project planning is beneficial to project performance.

While cause and effect relationships, and the degree of variance among variables, was not always clear and is still being studied, we could determine which measures seem to have the greatest influence on project performance and which ones are less important. In fact, using a filter of $p \leq 0.05$, we concluded that only 38 of the 83 measures examined (45%) seem to have statistical significance to project performance. It might be interesting to the readers that factors such as project complexity, project size, organizational structure and economic environment produced very low association levels to project performance and were among the 55% of the factors which were eliminated from further consideration.

For the current study, we grouped related measures to create the eight classes of "conditions" as described in Table I. Each class represents an aggregated set of measures which, based on field studies conducted so far, represents a favorable association with project performance. We have labeled these sets of measures MCP's as they critically influence the team environment toward project performance and ultimate project success. Subsequently, the MCP's, together with the descriptions shown in Table I, have been used to form the questionnaire for the current study, measuring the perceived strength and importance of each MCP within each project organization of our sample, as discussed in Section IV of this paper.

IX. STATEMENT OF USEFULNESS TO TRANSACTIONS READERS

The results of a comparative field study, conducted in Brazil and the United States, show that successful integration of multinational projects may not require fundamentally different management approaches, but it requires strong senior management support and efforts of developing effective organizational linkages and alliances with focus on cross-boundary relationships, negotiations, delegation, and commitment. The findings presented in this paper should help to stimulate further managerial thought and insight for both management scholars and practitioners toward more comprehensive models and management tools in order to achieve the high performance needed in today's technology-oriented multinational project environments.

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